Que. $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2+y^2+z^2}} = \frac{dt}{t - az}$

$\log x = \log y + \log c_1$

$\boxed{\frac{x}{y} = c_1}$

$x^2 + y^2 + z^2 = t^2$

$x\,dx + y\,dy + z\,dz = t\,dt$

$x, y, z \rightarrow$ multiplier

$$\frac{x\,dx + y\,dy + z\,dz}{x^2+y^2+z^2 - az\sqrt{x^2+y^2+z^2}} \Rightarrow \frac{\cancel{t}\,dt}{t^{\cancel{2}} - azt} \Rightarrow \frac{dt}{t - az}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - at} = \frac{dt}{t - az}$$

(i) (ii) (iii) (iv)

add (iii) + (iv):- $\frac{dz + dt}{z - at + t - az} = \frac{dz + dt}{z + t - a(z+t)}$

$$= \frac{dz + dt}{(z+t)(1-a)}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2+y^2+z^2}} = \frac{\cancel{dt}}{t - az} = \frac{dz + dt}{(z+t)(1-a)}$$

I and v

$\frac{dx}{x} = \frac{dz + dt}{(1-a)(z+t)} \Rightarrow \log x = \log(z+t) + \log c_2$;

$x^{1-n} = (z+t)c_2 \Rightarrow \boxed{c_2 = \frac{x^{1-a}}{z+t}}$

Lagrange's form :-

$P_p + Q_q = R \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$f(c_1, c_2) = 0$

Imp Que

$p + 3q = 5z + \tan(y - 3x) \rightarrow$ Lagrange's

$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)} \rightarrow$ simultaneous

$\therefore P_p + Q_y = R$

$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ $c_1 =$ $c_2 =$ for $(c_1, c_2) = 0$ Answer

Solving further by

taking one variable const.

taking x = const. → $dx = 0$

$0 + dy + dz = 0$

Integrating → $\boxed{y + z = C} = f(x)$ — (2)

$\boxed{dy + dz = f'(x)}$ — (3)

From eqⁿ (1) → $dy + dz = -(y+z)\,dx$. — (4)

comparing eqⁿ (3) & (4) :- $-(y+z)\,dx = f'(x)$ — (5)

eqⁿ (2) → $y + z = f(x)$

$-f(x) = f'(x)$

$f'(x) + f(x) = 0$, or $\frac{d}{dx} f(x) + f(x) = 0$

$(D+1)\,f(x) = 0$

AE :- $m + 1 = 0 \rightarrow m = -1$

$CF = C_1 e^{-x}$ ∴ $f(x) = Ce^{-x}$

$\boxed{y + z = Ce^{-x}}$ Ans

Que. $3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z})\,dz = 0$ — (1)

Condⁿ of integrability

$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$

$3x^2(0 + 3y^2) + 3y^2(-3x^2) + (-x^3 - y^3 - e^{2z})(0) = 0$

$9x^2y^2 - 9x^2y^2 = 0$

One variable const → z = const. → $dz = 0$

$3x^2 dx + 3y^2 dy = 0$ — (2)

Integrating $x^3 + y^3 = C = f(z)$ — (3)

$3x^2 + 3y^2 = f'(z)$ — (4)

diff variable

I & II $\quad 3dx = dy$

$0 = dy - 3dx$

$$\boxed{c_1 = y - 3x}$$

I & II

$$\frac{dx}{1} = \frac{dz}{5z + \tan c_1}$$

$$x = \frac{\log}{5}(5z + \tan c_1) + \frac{\log c_2'}{5}$$

$$5x = \log(5z + \tan c_1)c_2$$

$$e^{5x} = (5z + \tan c_1)c_2$$

$$\boxed{\frac{e^{5x}}{5z + \tan(y-3x)} = c_2}$$

$$f\left(y - 3x, \frac{e^{5x}}{5z + \tan(y-3x)}\right)$$

Total differential equation # $\boxed{Pdx + Qdy + Rdz = 0}$

Que $(2x^2 + 2xy + 2xz^2 + 1)dx + dy + zdz = 0$

Que $(y^2 + yz)dx + (xz + y^2)dy + (y^2 - xy)dz = 0$

Que $3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z})dz = 0$

M-1 → Inspection method

M-2 → Taking one variable constant.

M-3 → Method of auxiliary equation.

M-4 → Homogeneous

condition of integrability :→

$$Pdx + Qdy + Rdz = 0$$

$$\begin{matrix} P & Q & R \\ x & y & z \end{matrix} \Rightarrow P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$$

Que $(y+z)dx + dy + dz = 0$ — (1)

$$Pdx + Qdy + Rdz = 0$$

cond^n of integ. :

$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) \quad (y+z)(0) + 1(-1) + 1(1) = 0$

I & II $3dx = dy$

$0 = dy - 3dx$

$$\boxed{c_1 = y - 3x}$$

I & III

$$\frac{dx}{1} = \frac{dz}{5z + \tan c_1}$$

$$x = \frac{\log}{5}(5z + \tan c_1) + \frac{\log c_2'}{5}$$

$$5x = \log(5z + \tan c_1)c_2$$

$$e^{5x} = (5z + \tan c_1)c_2$$

$$\boxed{\frac{e^{5x}}{5z + \tan(y-3x)} = c_2}$$

$$f\left(y - 3x, \frac{e^{5x}}{5z + \tan(y-3x)}\right)$$

Total differential equation

$$\boxed{Pdx + Qdy + Rdz = 0}$$

Que $(2x^2 + 2xy + 2xz^2 + 1)dx + dy + zdz = 0$

Que $(y^2 + yz)dx + (xz + y^2)dy + (y^2 - xy)dz = 0$

Que $3x^2 dx + 3y^2 dy - (x^3 + y^3 + e^{2z})dz = 0$

M-1 → Inspection method

M-2 → Taking one variable constant.

M-3 → Method of auxiliary equation.

M-4 → Homogeneous

Condition of integrability :→

$$Pdx + Qdy + Rdz = 0.$$

$$\begin{matrix} P & Q & R \\ x & y & z \end{matrix} \Rightarrow P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$$

Que. $(y + z)dx + dy + dz = 0$ — 1

$$Pdx + Qdy + Rdz = 0$$

cond^n of integ. :-

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) \quad (y+z)(0) + 1(-1) + 1(1) = \underline{0}$$

Maths

① Simultaneous differential equation.

Q1. $Dx - y = t$

$x + Dy = 1$

Cramer rule

$x = \frac{\Delta x}{\Delta}$ $\quad \Delta = \begin{vmatrix} D & -1 \\ 1 & D \end{vmatrix}$ $\quad \Delta x = \begin{vmatrix} t & -1 \\ 1 & D \end{vmatrix}$

$y = \frac{\Delta y}{\Delta}$ $\quad \Delta = D^2+1$ $\quad \Delta x = tD+1$

$z = \frac{\Delta z}{\Delta}$ $\quad \Delta y = \begin{vmatrix} D & t \\ 1 & 1 \end{vmatrix}$

$\Delta y = D - t$

$x = \frac{Dt+1}{D^2+1}$ $\quad y = \frac{D-t}{D^2+1}$

$(D^2+1)x = Dt+1$ $\quad (D^2+1)y = D-t$

$(D^2+1)x = 1+1$ $\quad (D^2+1)y = -t$ — (2)

$(D^2+1)x = 2$ — (1)

A.Equation

$m^2+1=0$

$m^2 = -1 \Rightarrow m = \pm i$

$CF = e^{0t}[c_1\cos t + c_2 \sin t]$

$PI = \frac{2}{D^2+1} = (1+D^2)^{-1}2 \Rightarrow (1-D^2+--)2 = \underline{\underline{2}}.$

$PI = \frac{-t}{D^2+1}$

$-(1+D^2)^{-1}t$

$\Rightarrow -[1-D^2+-]t$

$\Rightarrow -[t] = -t$

$x = CF + PI$

$= c_1\cos t + c_2 \sin t + 2.$

$y = CF + PI$

$= c_1 \cos t + c_2 \sin t - t.$

II form. $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Q. $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$

$\frac{dx}{y} = \frac{dy}{x}$

$x\,dx = y\,dy$

$\frac{x^2}{2} = \frac{y^2}{2} + \frac{2c}{2}$

$x^2 = y^2 + c_1$ — (1)

$\frac{dy}{x} = \frac{dz}{z}$

$\frac{dy}{\sqrt{y^2+c_1}} = \frac{dz}{z}$

$\log(y + \sqrt{y^2+c_1}) = \log z + \log c_2$

$y + \sqrt{y^2+c_1} = c_2 z$ — (2)

(1) and (2) are required result.

Q2. $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{dz}$

$\frac{dx}{x} = \frac{dy}{y}$

$\log x = \log y + \log c_1$

$\log x = \log y c_1$

$x = y c_1$ — (1)

$\frac{dy}{y} = \frac{dz}{z}$

$\log y = \log z + \log c_2$

$\log y = \log z c_2$

$y = z c_2$ — (2)

(1) and (2) are required solution

2023.

$\frac{dx}{ny - mz} = \frac{dy}{lz - nx} = \frac{dz}{mx - ly}$

II Total differential equation

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Differential equation

Simultaneous diff. eq^n :-

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Que: Solve:- $\frac{dx}{dt} = ny - mz$; $\frac{dy}{dt} = lz - nx$; $\frac{dz}{dt} = mx - ly$

Sol^n $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = dt$

$$\frac{dx}{ny - mz} = \frac{dy}{lz - nx} = \frac{dz}{mx - ly} = dt$$

Using multiplier method (denominator = 0)

$x, y, z \rightarrow$ multiplier

$$\frac{xdx + ydy + zdz}{nxy - mxz + lzy - nxy + mxz - lyz} = 0$$

$$xdx + ydy + zdz = 0$$

Integrating:

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C' \Rightarrow \boxed{x^2 + y^2 + z^2 = C_1} - ①$$

$l, m, n \rightarrow$ multiplier $\rightarrow \boxed{lx + my + nz = C_2} - ②$

Que Solve $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$. $\left[\begin{array}{l} x^2 - y^2 = C_1 \\ y + \sqrt{y^2 + C_1} = z\,C_2 \end{array}\right]$

Que Method of multiplier :

$$\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2 + y^2}$$

Sol^n - Multiplier $\rightarrow$ $y, x, -z$. $\Rightarrow 2xy - z^2 = C_1$

$\rightarrow x, -y, -z \Rightarrow x^2 - y^2 - z^2 = C_2$

Maths

Qu $Dx - 7x + y = 0$ (1)

$Dy - 2x - 5y = 0$ (2)

$m = \text{complex} = 6 \pm i.$

$x = e^{6t}(c_1 \cos t + c_2 \sin t)$ .. Ans

$y = e^{6t}\{c_3 \cos t + c_4 \sin t)$

Que solve simultaneous diff. eq^n:

$$t\frac{dx}{dt} = t - 2x.$$

$$t\frac{dy}{dt} = tx + ty + 2x - t$$

Sol^n

$$\frac{dx}{dt} + \frac{2}{t}x = 1.$$

Linear diff. eq^n: $IF = e^{\int \frac{2}{t}dt} = e^{2\log t}$

$$IF = e^{\log t^2} = \boxed{t^2 = IF}$$

$$t^2 x = \int t^2 dt$$

$$t^2 x = \frac{t^3}{3} + c. \Rightarrow \boxed{x = \frac{t}{3} + \frac{c}{t^2}}$$

eq^n (2): $\cancel{t}\frac{dy}{dt} = \cancel{t}x + \cancel{t}y - \cancel{t}\frac{dx}{dt}$ (by eq^n (1))

$$\frac{dy}{dt} = x + y - \frac{dx}{dt} \Rightarrow \boxed{\frac{dx}{dt} + \frac{dy}{dt} = x + y,}$$

$$\frac{dx + dy}{x + y} = dt \Rightarrow \log(x+y) = t + \log c_2$$

$$\log\frac{x+y}{c_2} = t \Rightarrow \frac{x+y}{c_2} = e^t \Rightarrow x + y = c_2 e^t$$

$$\boxed{y = c_2 e^t - x} \quad \boxed{y = c_2 e^t - \frac{t}{3} - \frac{c_1}{t^2}} \text{ Ans}$$

$\log \frac{I}{(y+z)} = \log c \Rightarrow I = c(y+z)^2$

$Px + Qy + Rz = c(y+z)^2 \Rightarrow y(x+z)\cancel{(y+z)} = c(y+z)^{\cancel{2}}$

$\boxed{y(x+z) = c(y+z)}$, Ans

Que $(yz+z^2)dx - (xz)dy + (xy)dz = 0$ — (1)

Soln condn of integrability :-

$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$

$= (yz+z^2)[-x-x] + (-xz)[y-(y+2z)] + (xy)[z-(-z)]$

$= (yz+z^2)(-2x) - xz(2z) + 2xyz = -2xyz + 2xz^2 - 2xz^2 + 2xyz$

$= 0.$

condn of integrability is satisfied.

$Px + Qy + Rz$

$= xyz + xz^2 - xyz + xyz = xyz + xz^2 \neq 0$

$I = xyz + xz^2 \quad = \cancel{x(yz+z^2)}$

$I = x(yz+z^2)$

$\frac{\text{Given eq}^n}{I} = \frac{(yz+z^2)dx - (xz)dy + (xy)dz}{xyz + xz^2}$ — (3)

~~$dI = dx(yz+z^2) + x(d(yz)+z^2)$~~

$dI - 2xz\,dy - 2xz\,dz$ - - -

$\frac{dI}{I} - \frac{2xz(dy+dz)}{xz(y+z)} \Rightarrow \log I - 2\log(y+z) = \log c$

$I = c(y+z)^2$

$xz\cancel{(y+z)} = c(y+z)^{\cancel{2}}$

$\boxed{xy = c(y+z)}$, Ans.

$$= \frac{(y^2+yz)\,dx + (xz+z^2)\,dy + (y^2-xy)\,dz}{y(x+z)(y+z)} = \frac{dI - \text{something}}{I} \quad \text{--- } (3)$$

$I = y(x+z)(y+z)$

dI = Given eqⁿ + something → dI − something = given eqⁿ

$dI = dy(x+z)(y+z) + y(dx+dz)(y+z) + y(x+z)(dy+dz)$

$dI = xy\,dy + xz\,dy + yz\,dy + z^2dy + (y^2+yz)(dx+dz) +$
$(xy+yz)(dy+dz)$ — (4)

$dI = xy\,dy + xz\,dy + yz\,dy + z^2dy + y^2dx + y^2dz +$
$yz\,dx + yz\,dz + xy\,dy + xy\,dz + yz\,dy + yz\,dz -$
$xy\,dz + xy\,dy.$

Given eqⁿ → $(y^2+yz)dx + (xz+z^2)\,dy + (y^2-xy)\,dz$

$$= \underset{(1)}{y^2dx} + \underset{(2)}{yz\,dx} + \underset{(3)}{xz\,dy} + \underset{(4)}{z^2dy} + \underset{(5)}{y^2dz} - \underset{(6)}{xy\,dz}$$

$$dI = \overset{(1)}{y^2dx} + \overset{(2)}{yz\,dx} + \overset{(3)}{xz\,dy} + \overset{(4)}{z^2dy} + \overset{(5)}{y^2dz} - \overset{(6)}{xy\,dz} +$$
$$[\underset{(7)}{2xy\,dy} + \underset{(8)}{2yz\,dy} + \underset{(9)}{2yz\,dz} + \underset{(10)}{2xy\,dy}]$$

$dI - [(7)\ (8)\ (9)\ (10)] =$

$dI - 2xy\,dy - 2yz\,dy - 2yz\,dz - 2xy\,dz$ = given eqⁿ

$dI - 2y(x+z)\,dy - 2y(x+z)\,dz$ = given eqⁿ

$$\frac{\text{eq}^n}{I} = \frac{dI - 2y(x+z)\,dy + 2y(x+z)\,dz}{I}$$

$$\frac{dI}{I} - \frac{(2y(x+z)\,dy - 2y(x+z)\,dz)}{I}$$

$$\frac{dI}{I} - \frac{2y(x+z)(dy+dz)}{y(x+z)(y+z)} = 0$$

$\log I - 2\log(y+z) = \log c.$

from eq(1) → $3x^2dx + 3y^2dy = (x^3+y^3+e^{2z})dz$ (5)

from (4) & (5) $(x^3+y^3+e^{2z})dz = f'(z)dz$

$f(z) + e^{2z} = f'(z) \Rightarrow \boxed{f'(z) - f(z) = e^{2z}}$

$IF = e^{\int -1 dz} = e^{-z}$

$\Rightarrow (IF)f(z) = \int e^{2z}\, IF\, dz \Rightarrow e^{-z}f(z) = \int e^{2z}e^{-z}dz$

$e^{-z}f(z) = e^{z} + C \Rightarrow f(z)e^{-z} - e^{z} = C$

$\boxed{(x^3+y^3)e^{-z} - e^{z} = C}$ Ans

$x^3 + y^3 = e^{2z} + Ce^{z}$ Ans

$\cancel{x^3+y^3 = e^{2z}+Ce^{2z}}$

Imp

Que: pay eqn → $(\cos x + e^x y)dx + (e^x + e^y z)dy + e^y dz = 0$ — (1)

Sol^n $\cancel{(\cos x + e^x y)[e^y z + e^y - e^y] + (e^x + e^y z)[0] + e^y[\ldots]}$

Cond^n

$(\cos x + e^x y)[e^y - e^y] + 0 + e^y[e^x - e^x] = 0$

One variable const. → $z = \text{Const} \rightarrow dz = 0$

$(\cos x + e^x y)dx + (e^x + e^y z)dy = 0$

Integrating

$\cancel{\sin x + ye^x + e^x y + ze^y = 0}$

$\Rightarrow \cancel{\sin x + 2ye^x + ze^y = 0}$

$\cos x\, dx + e^x y\, dx + e^x dy + e^y z\, dy = 0$

$\cos x\, dx + d(e^x y) + e^y z\, dy = 0$

$\sin x + e^x y + e^y z = f(z)$ — (2)

diff. → $\cos x\, dx + e^x y\, dx + e^x dy + e^y z\, dy + e^y dz = f'(z)dz$ (3)

$\cos x\, dx + e^x y\, dx + e^x dy + e^y z\, dy + e^y dz = 0$ (1)

comparing (1) & (3) → $f'(z)dz = 0 \Rightarrow f(z) = C$

From eq (2) ⇒ $\boxed{\sin x + e^x y + e^y z = C}$ Ans

Que: $(e^x y + e^z)dx + (e^y z + e^x)dy + (e^y - e^x y - e^y z)dz = 0$

Soln: cond^n:

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0$$

$\cancel{(e^x y + e^z)}[e^y \cancel{-(e^y - ze^y}$

$$(e^x y + e^z)[e^y - (e^y - e^x - ze^y)] + (e^y z + e^x)[-ye^x - e^z] + (e^y - e^x y - e^y z)[\cancel{e^x} - \cancel{e^x}]$$

$$= e^x y e^x + e^x y z e^y + e^z e^x + e^z z e^y - e^y e^x z y - e^y e^z z - e^x e^x y - e^x e^z$$

$= 0$

One variable const: $z = \text{const}$, $\to dz = 0$.

$$(e^x y + e^z)dx + (e^y z + e^x)dy = 0$$

Integrating: → $e^x y\,dx + e^z dx + e^y z\,dy + e^x dy = 0$

$$d(e^x y) + e^z dx + e^y z\,dy = 0$$

$$e^x y + e^z x + z e^y = f(z) \quad \text{(1)}$$

diff. →

Que → $p + 3q = 5z + \tan(y - 3x)$

Que:→ $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{x^2+y^2+z^2}}$

Que: $t\frac{dx}{dt} = t - 2x$

$t\frac{dy}{dt} = tx + ty + 2x - t$

Simultaneous diff. eq^n

Que Inspection method: ~~(yz+z~~

$(yz + 2x)dx + (xz - 2z)dy + (xy - 2y)dz = 0$

Sol^n cond^n of integrability is satisfied.

$$\begin{cases} ① \; d(xyz) = yz\,dx + xz\,dy + xy\,dz \\ ② \; d(x^2) = 2x\,dx \\ ③ \; d(yz) = z\,dy + y\,dz \end{cases}$$

Sol^n $d(xyz) + d(x^2) - 2d(yz) = 0$

$xyz + x^2 - 2yz = c.$

*Que ~~Ho~~ Method of homogeneous:

$(y^2 + yz)dx + (xz + z^2)dy + (y^2 - xy)dz = 0. \; —①$

Sol^n $Px + Qy + Rz \neq 0$

$xy^2 + xyz + \cancel{xyz} + yz^2 + zy^2 - \cancel{xyz} \neq 0$

$xy^2 + yz^2 + zy^2 + xyz \neq 0$

$I = y(y+z)(x+z) \neq 0. \; —②$

$\hookrightarrow \frac{1}{Px + Qy + Rz}$

→ $\frac{eq^n ① \text{ (given eq}^n)}{I \text{ (eq}^n ②)} = \frac{\text{Given eq}^n}{Px + Qy + Rz}$

$$\frac{dp}{-pq} = \frac{dq}{p^2} = \frac{dx}{-2py} = \frac{dy}{-2qy+z} = \frac{dz}{-2p^2y-2q^2y+z} \quad -(2)$$

$\Rightarrow \frac{dp}{-pq} = \frac{dq}{p^2} \Rightarrow \cancel{p\,dp = -q\,dq} \Rightarrow \frac{dp}{-q} = \frac{dq}{p} \Rightarrow p\,dp = -q\,dq$

$\frac{p^2}{2}\ \frac{p^2}{2} = -\frac{q^2}{2} + a \Rightarrow \boxed{p^2 + q^2 = a} \quad -(3)$

$\cancel{dz = p\,dx + q\,dy}$ put eqⁿ (3) in eqⁿ (1)

$\Rightarrow$ eqⁿ $\rightarrow (p^2+q^2)y = qz \Rightarrow ay = qz$

$$q = \frac{ay}{z}$$

$p^2 + \frac{a^2y^2}{z^2} = a \Rightarrow p^2z^2 + a^2y^2 = az^2 \Rightarrow p^2z^2 = az^2 - a^2y^2$

$p^2 z^2 = \frac{az^2}{z^2} - \frac{a^2y^2}{z^2} \Rightarrow \cancel{p = \sqrt{a(1-ay^2}}$

$$p = \sqrt{a - \frac{a^2y^2}{z^2}} = \sqrt{a} - \frac{ay}{z}$$

$dz = p\,dx + q\,dy \Rightarrow dz = \left(\sqrt{a} - \frac{ay}{z}\right)dx + \frac{ay}{z}dy.$

Que. $z^2dx + (z^2 - 2yz)dy + (2y^2 - yz - xz)dz = 0$ — (1)

Solⁿ cond^n of integrability:-

$Px + Qy + Rz = 0$

$x = uz \quad y = vz$.

$Px + Qy + Rz$

$\hookrightarrow xz^2 + yz^2 - 2y^2z + 2y^2z - yz^2 - xz^2 = 0$.

Put $x = uz$ & $y = vz$.

$dx = zdu + udz$; $dy = zdu + vdz$.

$z^2dx + (z^2 - 2vz^2)dy + (2v^2z^2 - vz^2 - uz^2)dz = 0$

$z^2(zdu + udz) + (z^2 - 2vz^2)(zdu + vdz) + (2v^2z^2 - vz^2 - uz^2)dz = 0$

$\Rightarrow z^3du + \cancel{uz^2dz} + z^3du + \cancel{vz^2dz} - 2vz^3du - \cancel{2v^2z^2dz} +$

$\cancel{2v^2z^2dz} - \cancel{vz^2dz} - \cancel{uz^2dz} = 0$

$\Rightarrow 2z^3du - 2vz^3du = 0 \Rightarrow zdu + zdv - 2vzdv = 0$

$du + dv - 2vdv = 0$

$u + v - v^2 = C \Rightarrow \frac{x}{z} + \frac{y}{z} - \frac{y^2}{z^2} = C$

$\Rightarrow \boxed{xz + yz - y^2 = Cz^2}$ Ans

Partial differential eqⁿ

$$\boxed{\frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}}}$$ — (2)

Charpit's Method

$f(x, y, z, p, q) = 0$ — (1)

$dz = pdx + qdy$ — (3)

Que:- $2xz - px^2 - 2qxy + pq = 0$ (1) solve by charpit's method

$$\frac{dp}{\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}} = \frac{dx}{\frac{-\partial f}{\partial p}} = \frac{dy}{\frac{-\partial f}{\partial q}} = \frac{dz}{-p\frac{\partial f}{\partial p} - q\frac{\partial f}{\partial q}}$$

$$\Rightarrow \frac{dp}{(2z - 2px - 2qy) + p(2x)} = \frac{dq}{(-2qx) + q(2x)} = \frac{dx}{-(-x^2+q)} =$$

$$\frac{dy}{-(-2xy+p)} = \frac{dz}{-p(-x^2+q) - q(-2xy+p)}$$

$$\Rightarrow \frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dx}{x^2-q} = \frac{dy}{2xy-p} = \frac{dz}{px^2 + 2xyq - 2pq}$$

$dq = 0 \Rightarrow q = a$ (3) $\rightarrow$ put in eq (1):

$2xz - px^2 - 2axy + pa = 0 \Rightarrow p(a - x^2) = 2axy - 2xz$.

$$p = \frac{2x(ay-z)}{a-x^2} \Rightarrow$$

$dz = p\,dx + q\,dy \Rightarrow dz = \frac{2x(ay-z)}{a-x^2}dx + a\,dy$

$$dz - a\,dy = \frac{2x(ay-z)}{a-x^2}dx \Rightarrow dz - a\,dy = \frac{2x(z-ay)dx}{x^2-a}$$

$$\frac{dz - a\,dy}{z - ay} = \frac{2x\,dx}{x^2-a} \Rightarrow \log(z-ay) = \frac{dt}{t}$$

$x^2 - a = t$

$2x\,dx = dt$

$$\log(z-ay) = \log(x^2-a) + \log c$$

$$\boxed{z - ay = (x^2-a)c}$$ Ans

Que. $(p^2+q^2)y = qz$ (1)

Soln.

$yp^2 + yq^2 - qz = 0$ (1)

$$\frac{dp}{-pq} = \frac{dq}{p^2+q^2+q^2} = \frac{dx}{-2py} = \frac{dy}{-2qy+z} = \frac{dz}{-p(2py) - q(2qy-z)}$$